

- *Better citizen services:* It helps to provide citizens with easier access to connected government services through tracking, search, and conversational bots.

Open source platforms for generative AI

There are numerous open source LLMs available for generative AI. Here is a brief introduction to a few of them.

TensorFlow: This is Google's end-to-end library for performing low level and high-end frameworks for generative AI. It provides a robust platform for developing and deploying generative AI models that can create realistic and diverse content, such as images, text, audio, and video. It can handle deep neural networks for image recognition, handwritten image/writing classification, recurrent neural networks, NLP, word embeddings and PDE (partial differential equation).

Falcon LLM: Released in September 2023, this is a foundational large language model (LLM) with 180 billion parameters trained on 3500 billion tokens. It generates human-like text, translates languages, and answers questions.

OpenMMLab: This computer vision algorithm system is a multimodal advanced, generative, and intelligent creation toolbox. It provides high-quality libraries to reduce the difficulties in algorithm reimplementation. It creates efficient deployment toolchains targeting a variety of backends and devices. OpenMMLab has released 30+ vision libraries, implemented 300+ algorithms, and contains 2000+ pretrained models.

Llama-2: Meta AI developed Llama-2 in July 2023. This foundation language model is trained on a trillion tokens with publicly available datasets. It is optimised for natural language generation and programming.

Flan-T5: Released in October 2022 by Google, this open source LLM is

available for commercial usage. This collection of datasets is used to improve model performance. As many as 1.8K tasks were phrased as instructions and used to finetune the model.

Mistral 7B: Released in September 2023, this is a pretrained generative text model with 7 billion parameters. The model is used in mathematics, code generation, and reasoning. It is suitable for real-time applications where quick responses are essential. Mistral is multilingual, which makes it a flexible model for a range of applications and users.

PaLM2: Released in May 2023, this language model performs multilingual and reasoning capabilities.

Code Llama: Released in August 2023 by Meta, it is designed for general code synthesis and understanding. It's a 34 billion parameter language model.

Phi-2: Released by Microsoft Research Lab in June 2023, this 2.7 billion parameter language model can be prompted using a QA format, a chat format, and the code format.

Bloom: Released in 2022, this is an autoregressive LLM that generates text by extending a prompt using large amounts of textual data. The model was trained on 176B parameters, and can write in 46 languages and 13 programming languages.

GPT-3 AI: This NLP model developed by OpenAI is a generative pretrained transformer that uses deep learning to generate human-like text. It can generate text without

any prior training or input. It is useful for a variety of tasks, such as natural language processing, machine translation, question answering, summarisation, etc. GPT-3 is highly scalable, meaning it can handle large amounts of data quickly and accurately.

Bert: Bert stands for Bidirectional Encoder Representations from Transformers. Released by Google in 2018, this 340 million parameter LLM leverages a transformer-based neural network to understand and generate human-like language. It performs various NLP tasks like task classification, question answering, and named entity recognition.

Other open source LLMs include OPT 175B, XGen-7B, GPT-NeoX, and Vicuna-13B.

In summary, it is very important to choose the right set of LLM models, architecture, and open source frameworks for generative AI for building industry solutions. Since there is so much choice, this can be a challenging task.

Benefits of open source LLMs include:

- Enhanced data security and privacy
- Better RoI
- Reduced vendor dependency
- Code transparency and language model customisation
- Active community support and fostering innovation

However, LLM technology is still quite new and not well understood yet. It continues to evolve. **END** 

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By: Dr Gopala Krishna Behara

The author is an enterprise architect in the BTIS Enterprise Architecture division of HCL Technologies Ltd. He has a total of 28 years of experience in the IT industry.

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